CONFORMATIONAL ISOMERISM.

Some important definitions

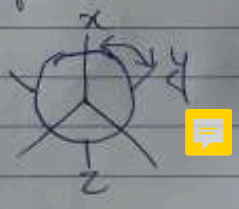
1) Conformations: Different 3-D arrangement of atoms or groups ~~which are interconvertible~~ of a compound which are interconvertible into each other at room temperature by rotation of σ bonds are called Conformations.

→ Stable conformations are called conformer = Rotomers = Conformational isomers.

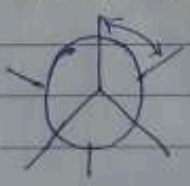
→ we can't seperate conformers.

2) Dihedral angle / Torsional angle: Angle between two bonds which are present of two adjacent atoms.

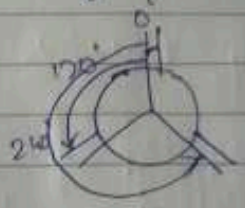
(angle of C - x and C - y in x - C - C - y)



3) Staggered Conformations: Conformations in which values of dihedral angle - 60°, 180°, 300°



4) Eclipsed conformation :-
Conformation in which value of dihedral angle = 0°, 120°, 240°



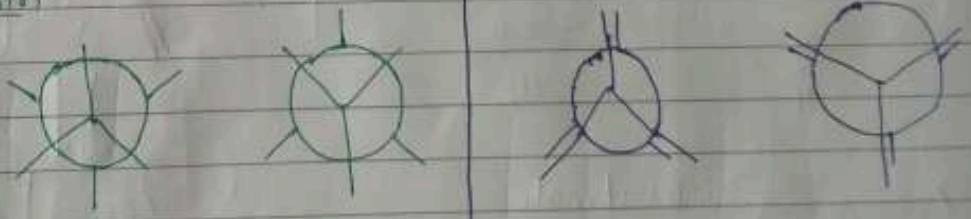
5) Skew configuration :- Conformations other than staggered and eclipsed are called skew conformation.

Representation of staggered and eclipsed

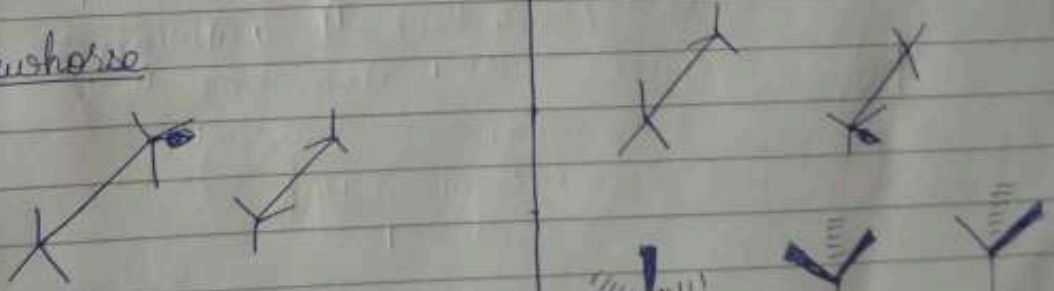
Staggered | Eclipsed

Newmann

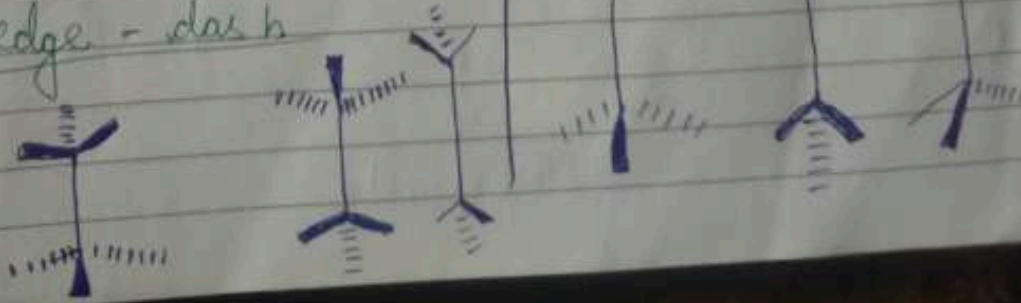
⇒



⇒ Sawhorse



⇒ wedge - dash



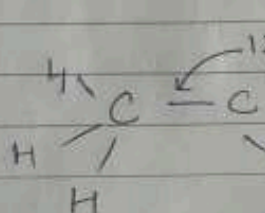
* We can't draw fischer structure of conformational isomerism

6) Energy barrier = Conformational energy

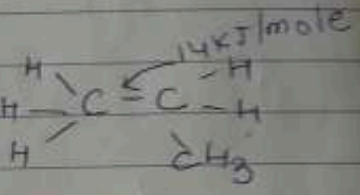
Energy required for complete rotation of a bond called conformational energy.

Energy difference between most stable and least stable conformation is called conformational energy.

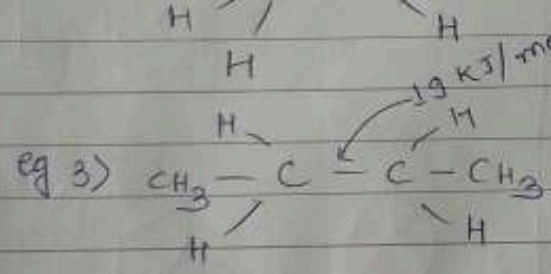
eg 1)



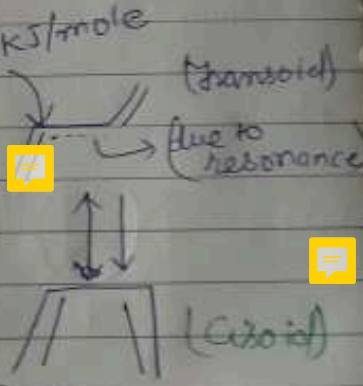
eg 2)



eg 3)



eg 4)

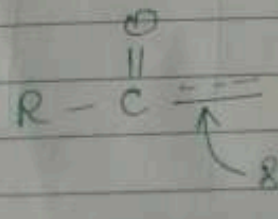


eg 5)



Room temp पे rotation नहीं होता। Room temp पे rotation करने के लिए 100 KJ/mol energy की आवश्यकता होती है।

eg 6)



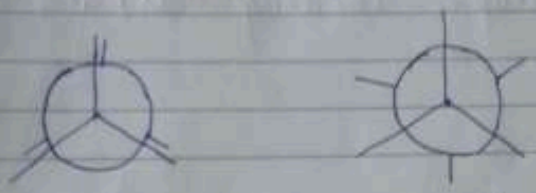
=> ~~energy barrier directly proportional to~~

=> energy barrier ∝ presence of bulky groups

" " ∝ multiple bond character

Factors affecting Stability of conformations

A) Torsional strain = Eclipsed strain



- Repulsion between electron density of two dihedral bonds.

=> Eclipsed > Staggered conformation (torsional strain)

=> Staggered structure are more stable than eclipsed structure.

B) Vander - waal's Strain

Repulsion b/w groups which are present on dihedral bonds.



C) Intramolecular hydrogen bonding

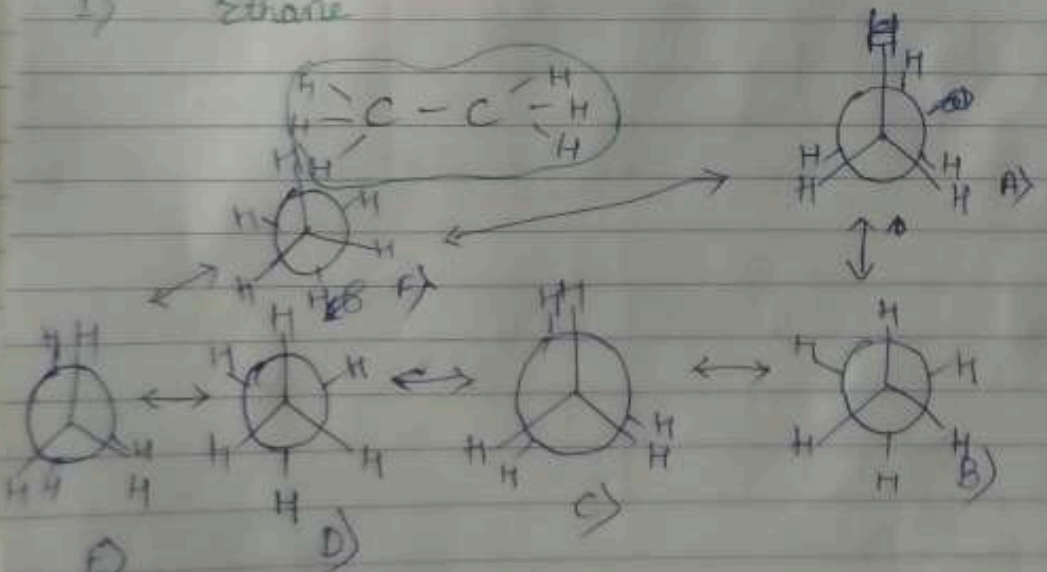
$-O \leftarrow H^{\delta+}$: F
$-N \leftarrow H^{\delta+}$: O
$-F \leftarrow H^{\delta+}$: N

* Staggered ~~structure~~ conformations which have intramolecular hydrogen bonding is/are most stable.

D) Gauche effect
E) Anomeric effect
F) Ring strain

Examples :-

1) Ethane

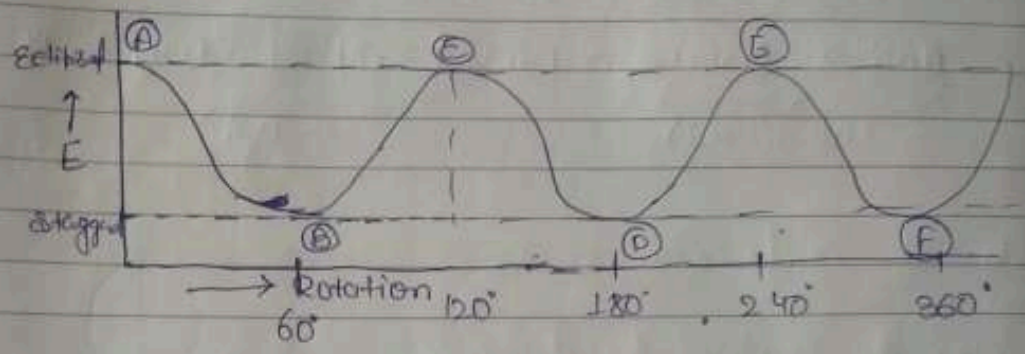


Stability ⇒ Staggered (~100%) > Eclipsed (~0%)

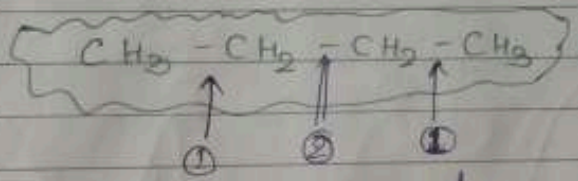
⇒ No. of Conformations = ∞

⇒ No. of conformational isomers = 3

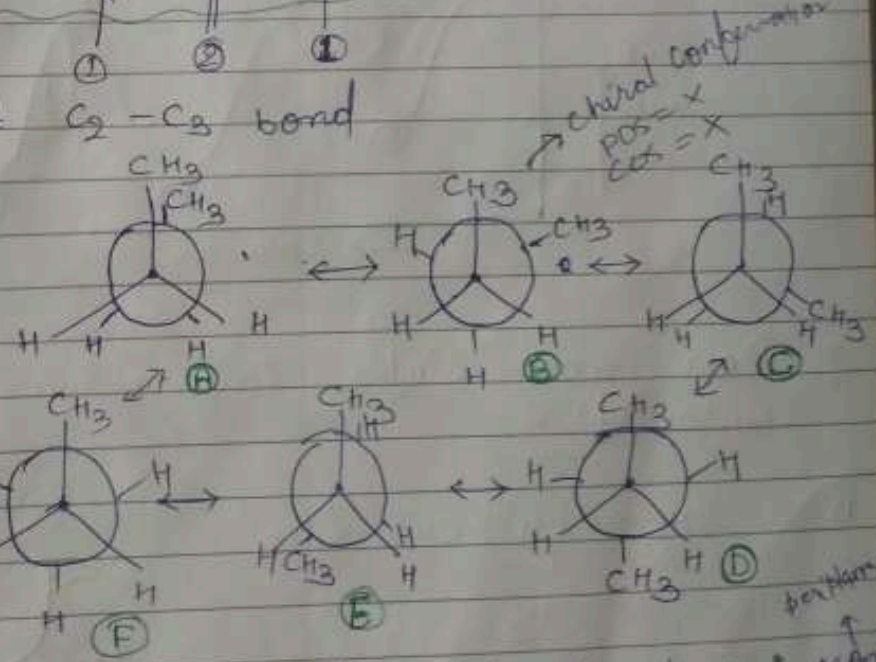
⇒ Energy barrier = 12 KJ/mole



2) Butane



⇒ w.r.t $C_2 - C_3$ bond



A) Fully eclipsed or Synperiplanar
B) Gauche structure
C) Partial eclipsed
D) ~~Partial eclipsed~~ Anti or Antiperiplanar
E) Partial eclipsed
F) Gauche

• Note: Structure (B) and Structure (D) are conformational enantiomers.

⇒ Stability of Conformational butane :-

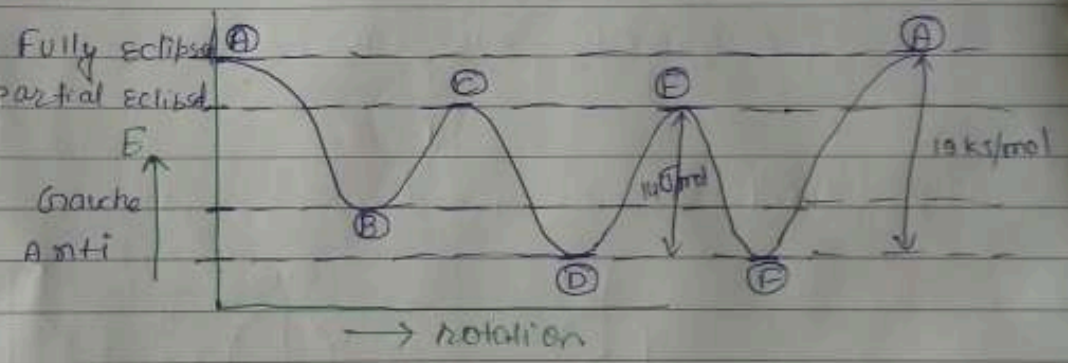
Anti > Gauche >>> partial eclipsed > Fully eclipsed

~100% (exist) (~0%)

⇒ No. of Conformation = ∞

⇒ No. of Conformational isomers = 3

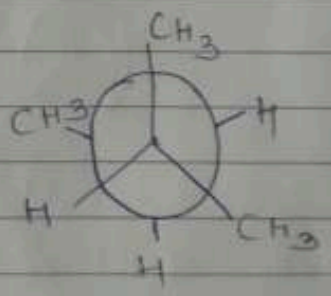
⇒ Energy graph for Conformation of butane.



Q. Draw most stable Conformation of given Compounds.

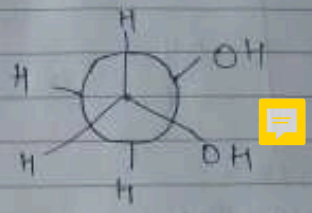
1) $CH_3 - \underset{\underset{CH_3}{|}}{CH} - CH_2 - CH_3$

$C_2 - C_3$



9)

$HO-CH_2-CH_2-OH$



यहाँ पर H bonding होता तो इसलिए ये ज्यादा stable है

यहाँ पर Gauche most stable than Anti

3)

3) $H-O-CH_2-CH_2-F$

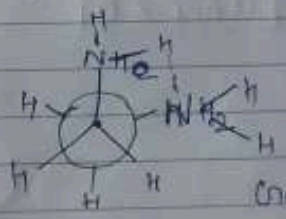
4) $H-O-CH_2-CH_2-CHO$

5) $H-O-CH_2-CH_2-COOH$

6) $H-O-CH_2-CH_2-NH_2$

Gauche. > Anti (Stability)

7) $\ddot{N}H_2-CH_2-CH_2-\ddot{N}H_2$



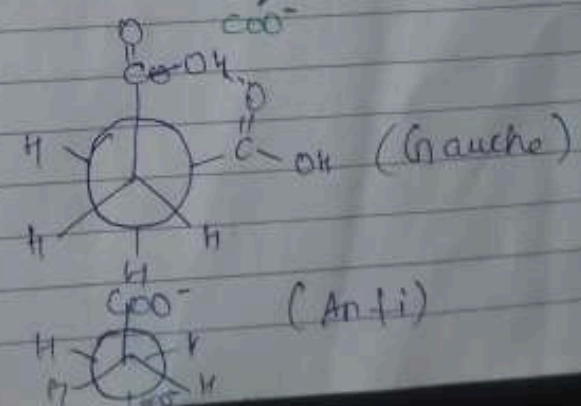
Gauche

8)

Succinic acid

in acidic med. → COOH, COOH

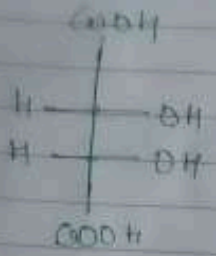
basic med. → COO^-, COO^-



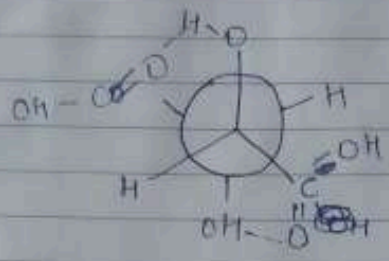
Acidic med. (Gauche)

basic (Anti)

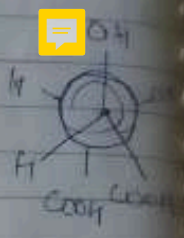
9)



(Meso tartaric acid)



(Anti)
(Most stable)

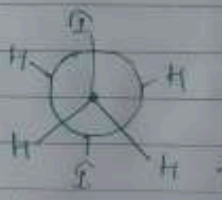


Gauche form
(less stable)

10)

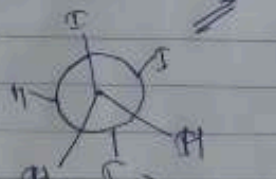
I – CH_2 – CH_2 – I

$\boxed{\mu_{net} \neq 0}$



$\mu_A = 0$

(Anti most stable)



(Gauche)

$\mu \neq 0$

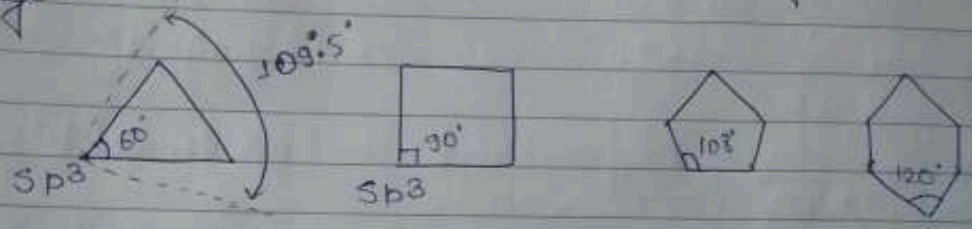
X_A

$\mu_{net} = \mu_A X_A + \mu_G X_G$

CONFORMATIONS OF CYCLOHEXANES

Baeyer's Ring strain :-

It is represented by difference b/w actual bond angle and ideal bond angle in rings.



ideal bond angle	= 109.5°	109.5°	109.5°	109.5°
actual "	= 60°	90°	108°	120°
(in planar)	49.5°	19.5°	1.5°	10.5°

theory के हिसाब से सबसे stable cyclopentane होना चाहिए पर experimentally यह cyclohexane सबसे ज्यादा stable होता है cyclohexane planner नहीं होता है

number of rings

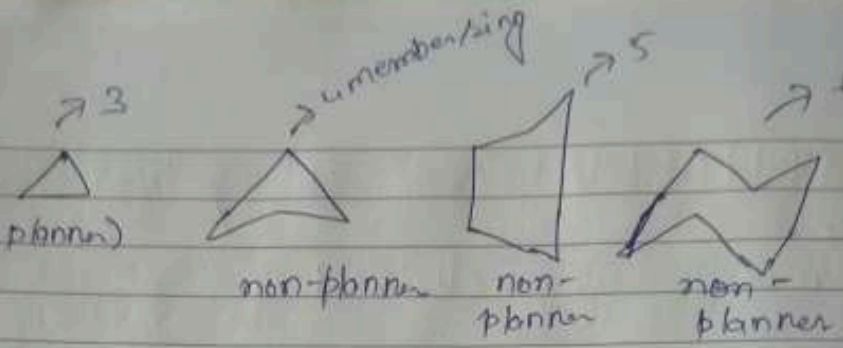
5 > 6 > 4 > 3 (X) Stability

6 > 5 > 4 > 3 > (✓) experimentally Stability

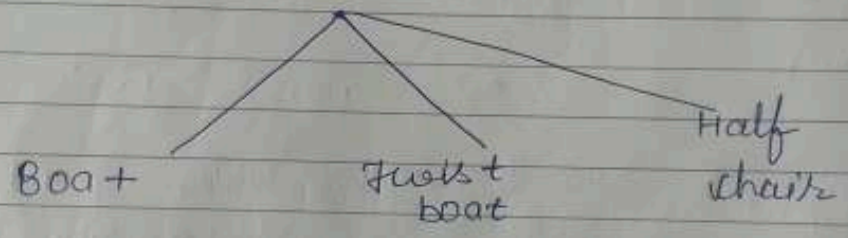
Stability exp :-

Cyclohexane > Cyclopentane > Cyclobutane > Cyclopropane

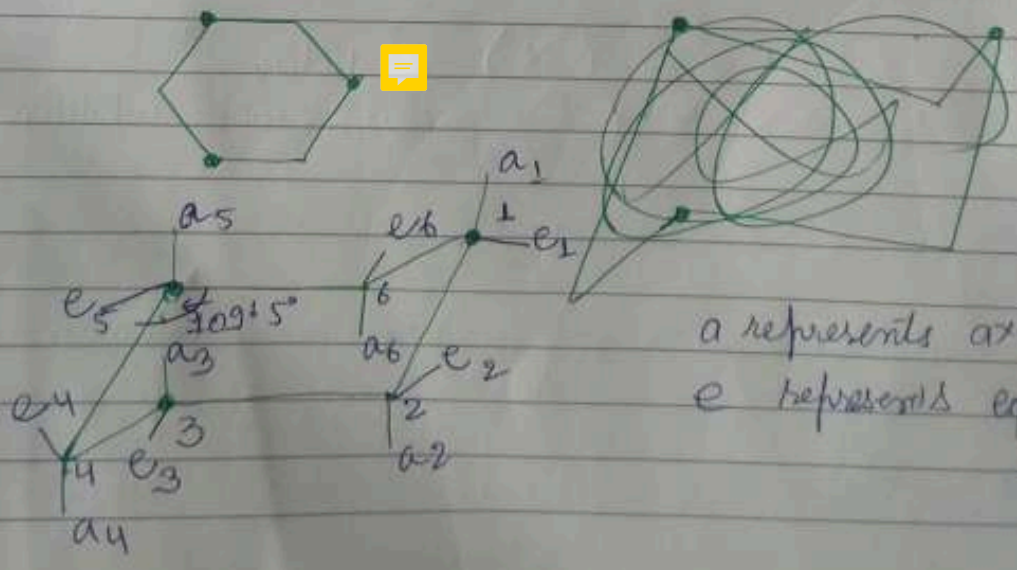
असल में सिर्फ 3 membered ring ही planner होता है,



- Cyclohexane is non-planner structure.
- It is exist in chair conformation. most imp
- The chair conformations are interconvertible into each other.
- Other conformations of cyclohexane are



A) Chair Conformation of Cyclohexane



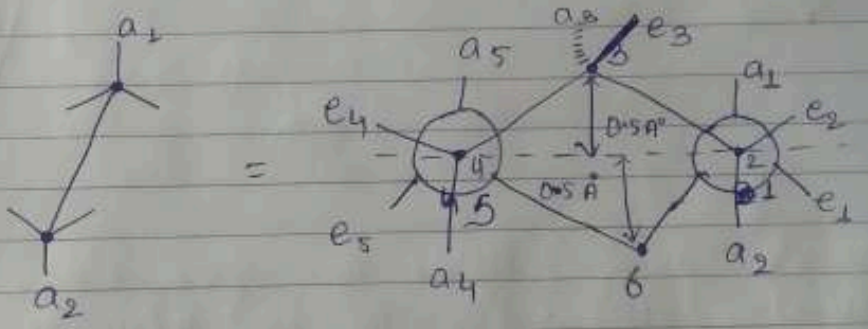
a represents axial
e represents equitorial

⇒ 6 axial position & 6 equitorial position.

⇒ $C_1 - C_2$ || $C_4 - C_5$ (bonds)
$C_2 - C_3$ || $C_5 - C_6$
$C_3 - C_4$ || $C_6 - C_1$

⇒ In cyclohexane (chair conformation), NO angle strain bcz bond angle = 109.5°

⇒ NO Torsional Strain (bcz it has staggered Conformation)



(Saw-horse) ↳ ☆ Newmann projection ≡ (chair form)

⇒ Identification of cis - trans

1) cis position w.r.t. a_1 = e_2, a_3 e_4, a_5, e_6

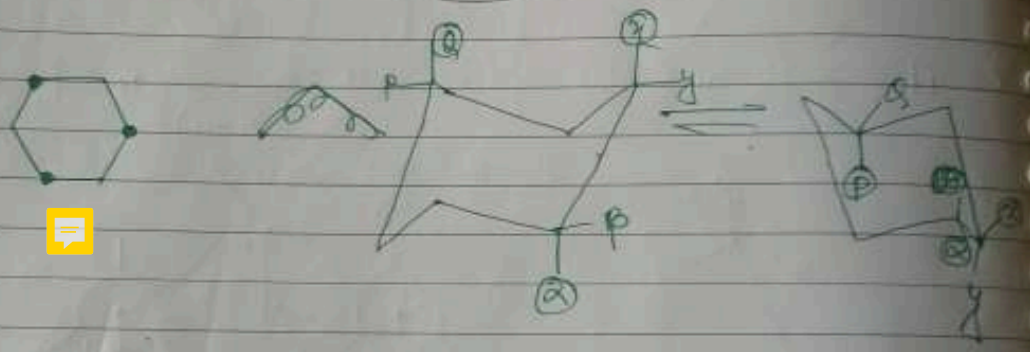
2) Transposition w.r.t. a_1 = a_2, e_3, a_4, a_6, e_5

3) cis " " " e_1 = a_2, e_3, a_4, a_6, e_5

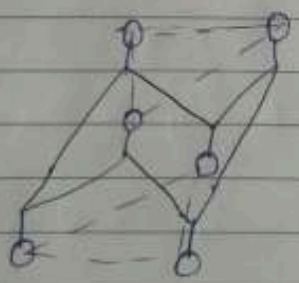
4) Trans " " " e_1 = e_2, a_3, e_4, a_5, e_6

- These chair conformations are interconvertible which is known as flipping in ~~chair~~ form Cyclohexane. and other conformations are form b/w two chair conformations.

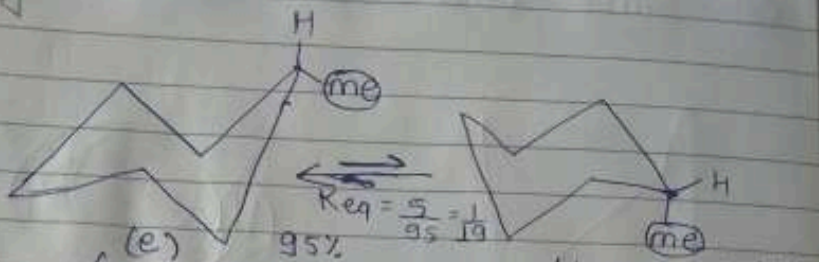
- → During flipping in cyclohexane (chair conformation) axial position and equitorial positions are interconvertible. (axial ⇄ equitorial)



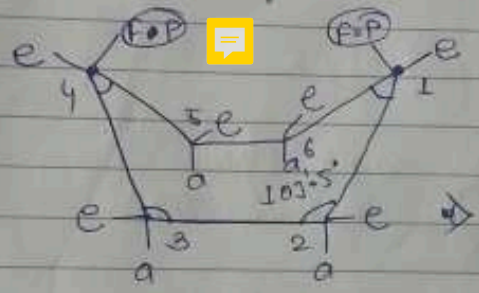
→ alternate axial positions are more closer than equitorial position. (a_1, a_3, a_5) & (a_2, a_4, a_6) so these axial positions are more steric hindrance position. Therefore bulky groups is more stable on equitorial position.



e.g Methylcyclohexane.



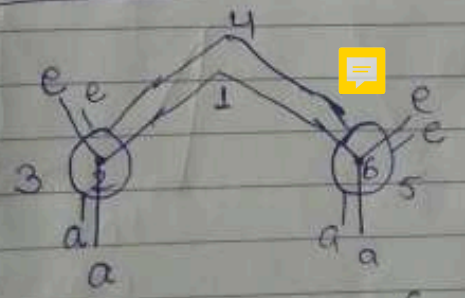
(B) Boat conformation :-



⇒ F.P = Flag pole positions

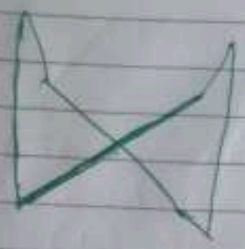
⇒ No angle strain (Bond angle = 109.5°)
⇒ It has maxm torsional strain bcz it has eclipsed conformation.

Newmann



⇒ Repulsion :- (F.P - F.P > axial - axial)

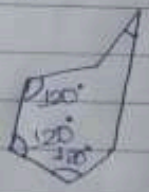
c) Twist boat conformation





⇒ partial angle strain
⇒ partial torsional strain

d) Half chair conformation

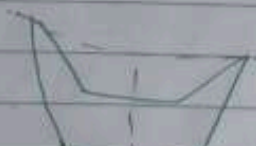


⇒ High angle strain
⇒ High torsional strain

Plane and Centre of Symmetry



POS = 3
COS = ✓



POS = 2
COS = x



* only conformation which is chiral.

POS = x
COS = x } chiral

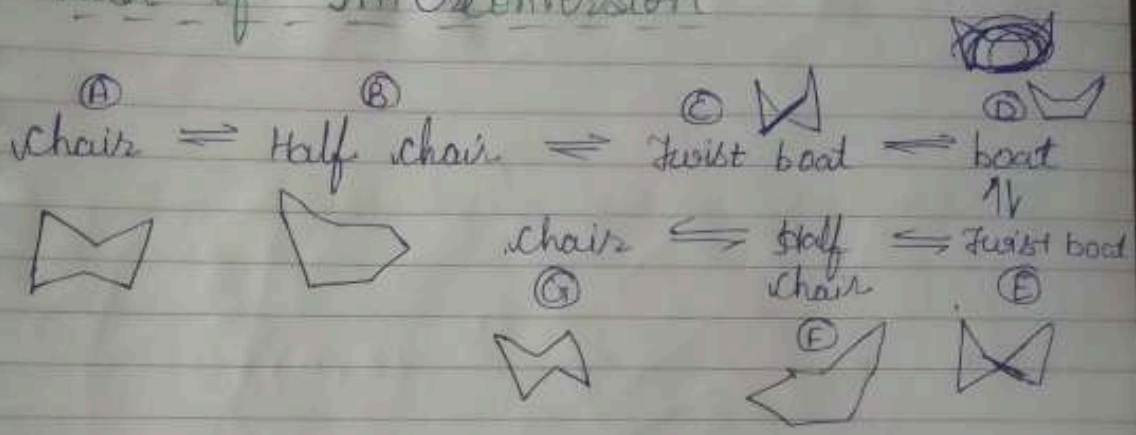


POS = ✓
COS = ✓

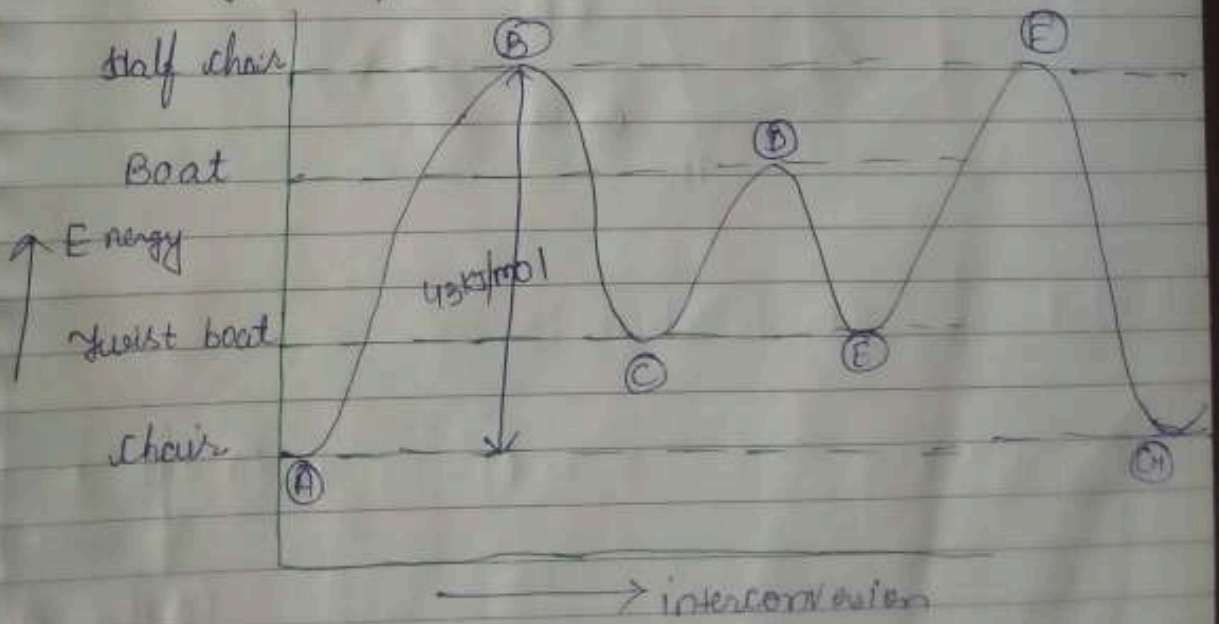
⇒ Stability order of Conformations of cyclohexane

Chair > Twist boat > Boat form > Half chair

⇒ Order of interconversion



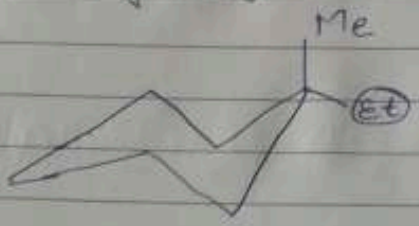
⇒ Energy profile



Q. Draw most stable conformation

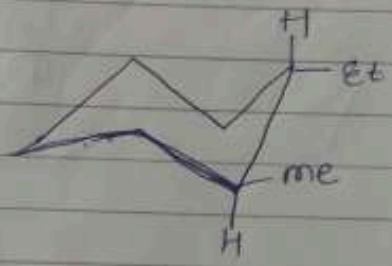
1) 1-Ethyl-1-methyl cyclohexane

⇒



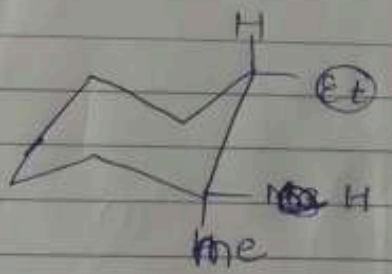
2) 1-Ethyl-2methyl cyclohexane

trans =



(e, e)

CIS



(e, a)

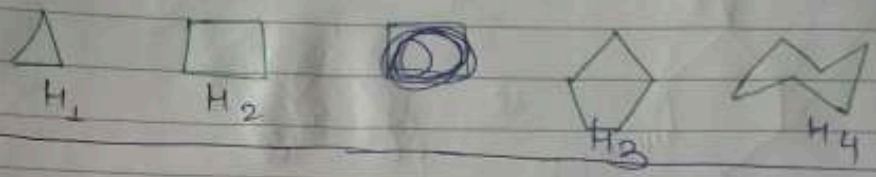
Heat of Combustion :-

$$C_x H_y + \frac{(2x+\frac{y}{2})}{2} O_2 \longrightarrow xCO_2 + \frac{y}{2} H_2O + \text{Heat of Comb.}$$

$$2x \qquad \frac{y}{2} \qquad (\Delta H_{comb})$$

⇒ Heat of Combustion ∝ no. of Carbon atoms

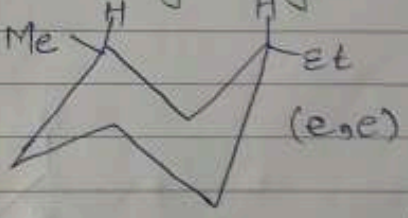
⇒ Heat of Comb. ∝ $\frac{1}{\text{Stability}}$ (for isomers)

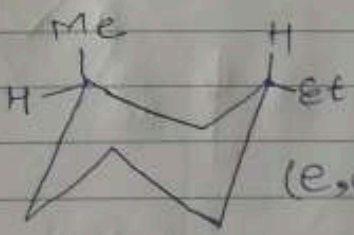
eg 

⇒ Heat of Comb. (per mole) = $H_4 > H_3 > H_2 > H_1$

⇒ Heat of Comb. (per CH_2) = $\frac{H_4}{6} < \frac{H_3}{5} < \frac{H_2}{4} < \frac{H_1}{3}$

3> 1 - Ethyl - 3 - methyl cyclohexane.

(3.a) cis ⇒ 

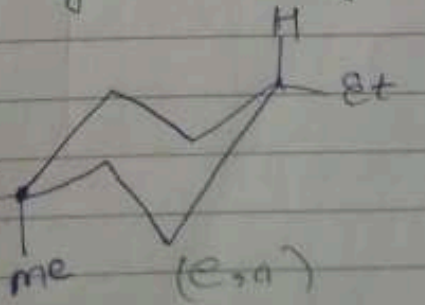
(3.b) Trans 

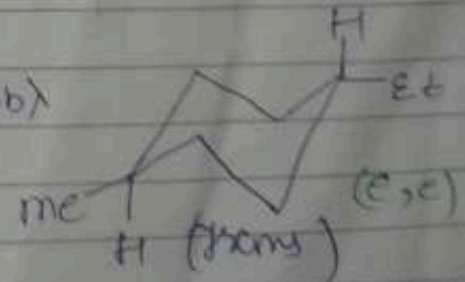
⇒ Stability ⇒ cis > Trans

⇒ Heat of Comb

⇒ cis < Trans

4> 1 - Ethyl - 4 - Methyl cyclohexane

(4.a) cis = 

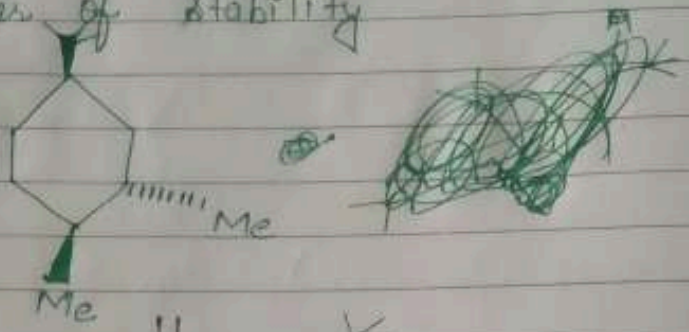
4.b) 

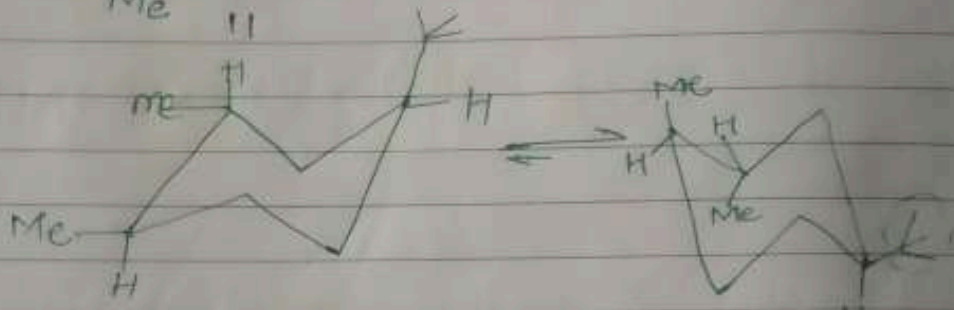
Stability = Trans > Cis

Heat of Combustion = Cis > Trans

Q. Give order of Stability

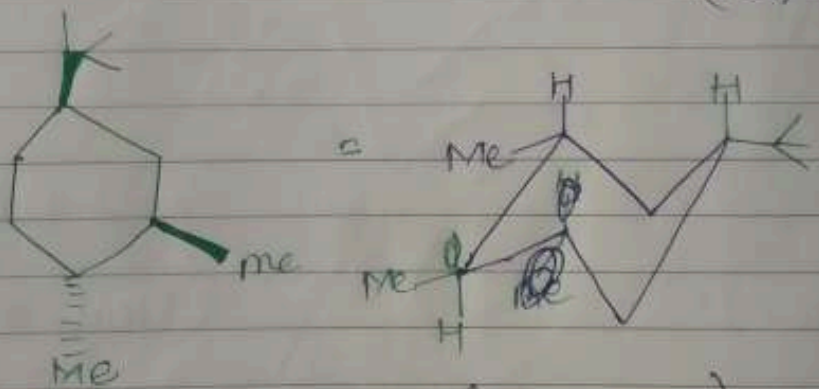
a)





(e, a, a)

b)



(e, e, e)
(More stable)

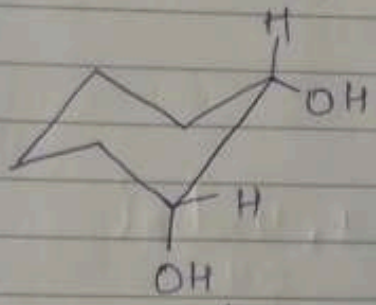
b > a

CONFORMATIONS Of Cyclohexane

Q. Draw most stable conformations :-

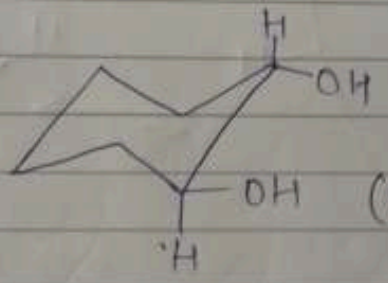
1) Cyclohexane 1,2-diol → cis, → trans

1.a) cis =



(e,a)

1.b) trans =

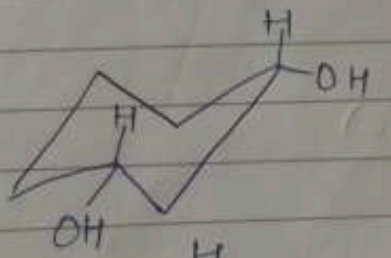


(e,e)

trans > cis

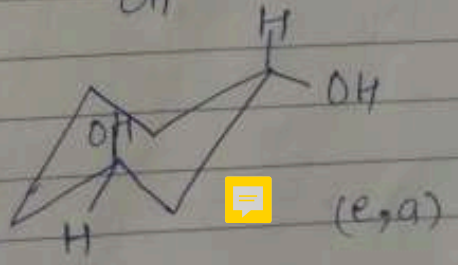
2) Cyclohexane-1-3-diol

⇒ cis



(e,e)

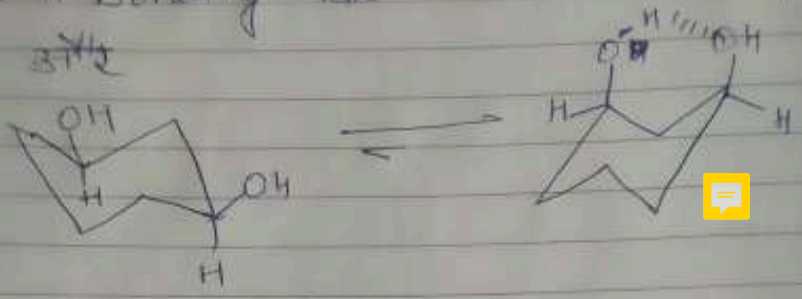
trans



(e,a)

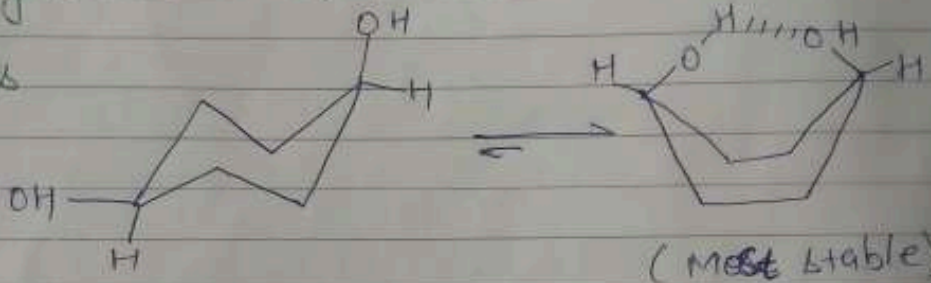
9-Aug-2020

due to H bonding cis ताने का flipping होता



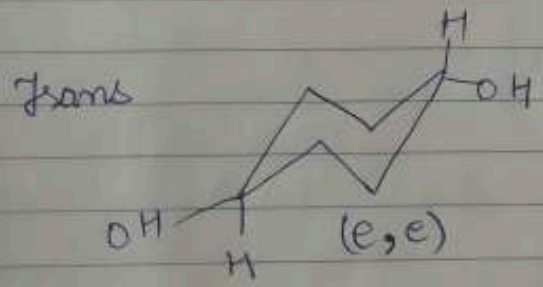
3) Cyclohexane 1, 4-diol

a) cis



(Most stable)

(Boat)



trans

(e, e)